1. Image Classification

Image classification means **predicting one label for the whole image**.

The model looks at the entire image and decides **which class it belongs to**.

Example

If the image contains a cat, the model output will be:

Cat

Even if there are multiple objects in the image, the model still gives **only one main label**.

Key Point

It tells **what is in the image**, but **not where it is located**.

Example Models

- ResNet
- VGG
- MobileNet

Example Applications

- Animal classification
- Plant disease detection
- Face recognition

Interview Line

Image classification is a task where a model predicts a single label for the entire image.

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2. Object Detection

Object detection means **detecting multiple objects in an image and locating them using bounding boxes**.

The model predicts:

1. **Object class**
2. **Object location (bounding box)**

Example

If the image has a **cat and a dog**, the output will be:

Cat → Bounding Box
Dog → Bounding Box

Key Point

It tells **what objects are present and where they are located**.

Example Models

- YOLO
- SSD
- Faster R-CNN

Example Applications

- Self-driving cars
- Security surveillance
- Traffic monitoring

Interview Line

Object detection identifies multiple objects in an image and predicts their locations using bounding boxes.

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3. Image Segmentation

Image segmentation means **classifying each pixel of the image**.

Instead of bounding boxes, it gives **precise object boundaries**.

Each pixel gets a class label.

Example

If the image contains:

- Car
- Road
- Person

The model labels pixels as:

Car pixels
Road pixels
Person pixels

Types of Segmentation

Semantic Segmentation

All objects of the same class are treated as one group.

Example:
All cars → same label

Instance Segmentation

Each object instance is identified separately.

Example:
Car 1 and Car 2 are different objects.

Example Models

- U-Net
- Mask R-CNN
- DeepLab

Example Applications

- Medical imaging
- Self-driving cars
- Satellite image analysis

Interview Line

Image segmentation classifies every pixel in an image to identify exact object boundaries.

Main Difference

Task	What it Predicts	Output Type
Image Classification	What object is in the image	One label
Object Detection	What objects and where	Bounding boxes
Image Segmentation	Exact object boundaries	Pixel-level labels

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1. Multi-Class Classification

Multi-class classification means the model chooses **only one class from multiple possible classes**.

So **one input → one output label**.

The classes are **mutually exclusive**, meaning only one can be correct.

Example

Suppose we have a model that classifies animals.

Possible classes:

- Cat
- Dog
- Horse

If we give an image of a dog, the output will be:

Dog

The model **cannot choose multiple classes**.

Example Applications

- Digit recognition (0–9)
- Animal classification
- Plant disease classification

Interview Definition

Multi-class classification is a problem where the model predicts one class from more than two possible classes.

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2. Multi-Label Classification

Multi-label classification means the model can predict **multiple labels for a single input**.

So **one input → multiple output labels**.

Classes are **not mutually exclusive**.

Example

Suppose an image contains:

- Dog
- Ball
- Grass

The model output can be:

Dog ✔
Ball ✔
Grass ✔

All three labels can be predicted at the same time.

Example Applications

- Image tagging (dog, park, tree)
- Movie genre prediction (action + comedy)
- Medical diagnosis (multiple diseases)

Interview Definition

Multi-label classification is a problem where one input can belong to multiple classes at the same time.

Main Difference Between Multi-Class and Multi-Label

Feature	Multi-Class	Multi-Label
Output labels	Only one label	Multiple labels
Class relationship	Mutually exclusive	Not exclusive
Example	Cat OR Dog OR Horse	Dog AND Ball AND Grass